

A Survey of Character Recognition in Tibetan Historical Documents

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ABSTRACT

Text recognition for Tibetan historical document images is the automated process of extracting and identifying text from these photos utilizing image processing, computer vision, and natural language processing techniques. It is generally understood to be a method of extracting text from visual data. In order to create editable, searchable, and analytical text forms that readers, researchers, and academics may study, preserve, and pass on, it attempts to analyze, identify, and change the text found in traditional Tibetan historical document images. Recent years have seen significant advancements in the effectiveness and performance of Tibetan historical document image and text identification thanks to the quick development of artificial intelligence technology. However, issues with low image quality, non-standard formatting, and notable font style inconsistencies persist in the field of image recognition of Tibetan historical document texts. As a result, the accuracy and universality of current recognizers are low. First, this article gives a general overview of the basic knowledge of Tibetan historical documents, which are the research objects in this field. This helps to better summarize the work of predecessors and support future research. The methods of integrating deep learning and conventional methods for Tibetan historical document image text recognition were then arranged, categorized, summarized, and introduced around the four subtasks of dataset construction, text detection, text recognition, and layout analysis. This served as the basis for sorting through the pertinent statistics and assessment indicators and summarizing the state and advancement of the research. Lastly, future technical development patterns are projected based on the primary issues and problems in this subject that require immediate attention.

Keywords: Tibetan historical documents; text detection; text recognition

1. INTRODUCTION

The process of automatically extracting and identifying text from Tibetan historical document images through the use of image processing, computer vision, and natural language processing technologies is known as text recognition of Tibetan historical documents. It may be roughly defined as a method of word proposal based on visual data. For academics, researchers, and readers to study, preserve, and transmit, it seeks to analyze, identify, and change text from conventional images of Tibetan historical writings into an editable, searchable, and analyzable digital form. Offline or offline documents are another term for papers that take the shape of images. Historically, offline character recognition, or optical character recognition (OCR), is an historical technology, but it has only happened in the past few years since it came out of the laboratory and into people's lives. Since 1998, Tibetan OCR has received extensive attention and research from the domestic academic community^[1]. In the past, the concept of Tibetan OCR was more limited to the recognition of modern printed Tibetan documents and then gradually broadened to general text and image recognition. Text recognition of Tibetan historical document images is more difficult than that of modern printed Tibetan document images because of the non-standard and complex layout, the deformation of the paper, the distinctive lettering and writing styles of the historicals in different periods, and the fading, staining, and blurred handwriting of the paper due to the erosion of time and improper preservation. Tibetan historical document recognition is frequently regarded as a distinct type of Tibetan OCR.

The text identification of images in Tibetan historical texts has evolved over the course of more than 30 years, encompassing a broad spectrum of techniques from statistical learning approaches and basic rule heuristics to deep learning methods. Early methods relied mostly on manual feature extraction and rule building, with basic image matching and template matching serving as the basis. These methods had stringent criteria for character image quality, low recognition accuracy, poor generalization ability, and limited processing effect for texts with varying styles and typefaces. With the advancement and use of deep learning technology in recent years, Tibetan history book intelligent text recognition performance has also significantly increased, and offline text recognition accuracy is now almost on par

with human levels. Numerous studies have been conducted on the picture text recognition technology of Tibetan historical documents, and several recognition systems have been developed, but there are still many issues that need to be resolved in practical application. This is due to advancements in related technologies and computer capacity. A few reviews of deep learning approaches have been published in the last several years in the fields of layout analysis^[2], text recognition^[3], and document intelligent analysis and recognition^[4]. However, because these evaluations focus on diverse recognition contexts, they are not very useful. Two reviews, titled "Layout analysis of Tibetan historical books in the era of deep learning"^[5] and "Text detection in the scene of Tibetan historical documents"^[6], have been published in the field of Tibetan historical text recognition. The goal of this literature review is to serve as a useful academic resource for academics or scholars who are just beginning to work on related tasks. It specifically addresses the problem of image and text identification of Tibetan historical texts.

2. OVERVIEW OF TIBETAN HISTORICAL DOCUMENTS

There are three basic categories for the Tibetan rendition of old literature (ལཱ་གཞི): etched, written, and printed. The "Meditation on the Merits of the Buddha Mother Dharani Sutra Strategy" was the first Tibetan inscription discovered in Heishui City during the Yuan Dynasty. The original format of Tibetan literature was known to have been created on paper scrolls, according to the Tibetan manuscript form discovered in Dunhuang. The first printed form of the Kangyur in Tibet is the Litang Zhu Plate, which signifies the advancement of Tibetan printing and engraving techniques^[8]. Manuscripts, stone inscriptions, and woodcut texts—of which there are the most—are the three subcategories into which some scholars have separated the editions of ancient Tibetan writings^[7]. The Longfu Temple edition in Beijing has the oldest woodcut, which is followed by the Natang edition in Shigatse, the Litang edition once more, and the most recent meditation edition by Gannan Zhuoni. It is evident from the pertinent research studies on the compilation of Tibetan literature that Tibetan historical books are bound in lengthy, line-bound, folded, and butterfly-shaped formats. The lengthy binding of Tibetan historical writings became the more stable format with the arrival of Buddhist texts.

The elements of the layout structure are as follows: a page center, a sidebar (མཐའ་ཐོག), a clear box (ཐུན་ཐུན), a boundary (ཐོག་ཐོག)^[8]. To ensure that the left and right ends of every page in the manuscript are aligned, a vertical line is drawn on both the left and right sides of the page. Alternatively, a vertical line is drawn to align the left and right ends of the picture and text. The Tibetan page number, title, volume label, etc. will be written vertically outside the left end boundary. Double vertical lines are occasionally drawn for artistic purposes as well. In general, there are two types of vertical lines: single vertical lines on the left and right, and double vertical lines that are thick on the outside and thin on the inside. However, there are other situations in which determining an individual's thickness is difficult. The line surrounding the frame content on each page is referred to as the sidebar or frame. It serves to standardize the layout and stops engraving, trimming, and binding from causing missing lines and words from a page. Depending on how many column lines there are, the sidebar can have four sides: left and right, higher and lower, four sides, etc. There are four perimeter fences: two fences around the perimeter known as bilateral sides; two lines on the left and right sidebars known as left and right bilateral; and one perimeter fence known as the surrounding unilateral line. Of course, some historical books have no sidebar at all, and the text ends on the left and right are almost exactly on the page. Mingmu is the term for the vertical grid, resembling the present bookmark, that is etched on the left and right sides of a lengthy page with the title, chapter page number, and Tibetan letters of the volume label. The vertical grid will always be at the left end of the front page of each page with the standardization of the content of the inscription of the historical Tibetan writings' purpose, but it will never be clearly apparent in the printed version. The centerpiece, which excludes the sidebar, the text, and the pictures of the Buddha, is a crucial component of the entire page. Figure 1 illustrates the layout structure of old Tibetan books. There is a significant amount of human tampering because historical accounts are written or carved differently. Each Tibetan historical book has its own distinct layout style, and the layout structure is not quite the same across the many copies.

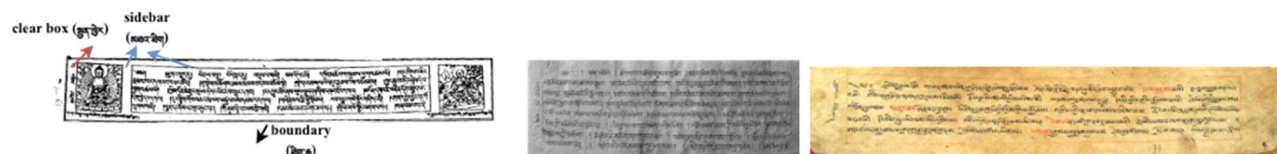


Figure 1. Layout structure of historical Tibetan books.

Tibetan syllables and Sanskrit-transcribed Tibetan syllables make up the Chinese version's composition. The fundamental distinction between the two is seen in the way Tibetan syllables are generally structured. The constituents of typical Tibetan syllables are vowel symbols, consonant letters, consonant letters up, consonant letters down, and consonant letters followed by consonant letters. Tibetan syllables transliterated into Sanskrit are represented by a vertical combination of the fundamental consonant letters, two consonant letters, and two vowel letters, like in ལྷོ་ལྷོ་. While some Sanskrit transcribed Tibetan letters have two vowels (the general structure of which is shown in GB/T 36335-2018), some Tibetan syllables have two lower consonants.

Researchers have discovered that the quantity and diversity of the extant Tibetan old writings are astounding based on their preliminary examination. For instance, since 1992, 1.2 million volumes of historical Tibetan literature have been published by a Tibetan computer firm in Nepal. The Latin alphabet is used by the Asian Classics Input Project "<http://buddhism-dict.net/ebti/textinput/acip.html>" to input Tibetan literature into a computer. 6,000 Tibetan Buddhist texts are available through the Tibetan Buddhist Resource Center (TBRC, "<https://tbcnetwork.org/>"). The Latse Modern Tibetan Library in the United States, the Himalayan Tibetan Digital Library at the University of Virginia in the United States, and the Digital Library for Tibetan Literature Research at Otani University in Japan are a few examples of comparable Tibetan literature databases. Nevertheless, the majority of these databases are kept as PDF files or images, which makes it difficult to mine literature for information, build knowledge bases, and create retrieval systems. In light of this, one of the key research questions in the field of digitizing Tibetan historical manuscripts is how to teach computers to automatically extract and detect text from the photographs of these documents.

3. OCR'S TECHNOLOGICAL ADVANCEMENTS IN TIBETAN HISTORICAL DOCUMENTS

Pattern recognition of Tibetan text in historical sources is an extremely difficult challenge. To be more precise, the model must identify and detect text in an input image of a Tibetan historical document in order to convert it into a digital format that can be edited, searched, and analyzed. Figure 2 provides an example in particular. Numerous techniques have been developed by researchers; they can be broadly categorized as deep learning techniques based on different deep neural networks and classic techniques based on pattern recognition and statistical learning.



Figure 2. Example of Tibetan text recognition task for images of historical documents.

3.1 Traditional math of Text recognition inTibetan historical documents

As illustrated in Figure 3, the conventional Tibetan recognition system for historical documents consists of six basic components: data preparation, layout segmentation, character segmentation, layout analysis, character recognition, and post-processing. In a nutshell, it is a segmentation strategy-based technique for Tibetan character identification.



Figure 3. Traditional Tibetan OCR recognition process.

Masami Kojima of the Tohoku Institute of Technology in Japan initiated research on character identification of photographs in Tibetan historical documents in 1991 [9]. In 2000, a character recognition system for Tibetan historical books was developed. Keutzer also recommended that master's student Fares Hedayati continue his research, and their work was published in Hedayati's master's thesis, "Recognition of Tibetan wood block prints with generalized hidden Markov and kernelized modified quadratic distance function." Later, in the late 1980s of the 20th century, Keutzer and Henry Baird collaborated to develop a comprehensive recognition scheme and applied the generalized hidden Markov model to Tibetan historical book character segmentation[10].

Almost all pattern classification techniques, which often involve binarization and tilt correction through image preprocessing, are the fundamental algorithms of traditional Tibetan historical text recognition research. In particular, a binarization technique based on Lab color space channel combination and local processing based on various Tibetan historical text backdrop colors was proposed by Han Yuehui et al. [11]. The same year, Gao Fei et al. introduced an image binarization approach based on historical book images that have been damaged. The method divides the picture of the deteriorated historical book text and backdrop and extracts the image's crucial value from the damaged historical book image. This allows for the adaptive compensation of the Tibetan historical book image [12]. For historical Tibetan woodcut images, a geometrically corrected-based tilt correction approach has been proposed [13], which is said to be more efficient than the projection contour method and the traditional Hough method. Because of the intricate design of Tibetan historical texts, the layout will first need to be separated or examined in order to accurately extract the text area. Published research work includes text region localization techniques based on bounding box correction algorithms and projection analysis based on connected domains and vertical horizontals [14]. Using the variations in the projection values of each layout structure element of the image of Tibetan historical documents following two segmentations and several projections, Wang Xiyou et al. were able to accomplish the task of segmenting Tibetan history books [15]. Ren et al. suggested a layout segmentation approach for Tibetan history books that classifies the feature vectors of Tibetan historical document images consisting of grayscale, color, and texture data using Support Vector Machines [16]. Text region in Tibetan historical document picture can be precisely determined through layout analysis; Tibetan characters in the text area must then be recognized; early text line and single character segmentation is required. According to Table 1, numerous scholars have suggested various techniques for precisely segmenting Tibetan characters because the majority of Tibetan historical records are handwritten and woodcut, which results in common character adhesion and overlap between the lines.

Table 1. Segmentation method of image adhesion text line segmentation of Tibetan historical documents.

Methods	Text line segmentation	Word segmentation
Text line baseline estimation or detection ^[17-24]	√	x
Text Line Baseline Estimation or Detection ^[25-27]	√	√
Connectivity domain analysis ^[28]	√	√
Connected domain analysis and A* algorithm ^[29]	√	x
Connectivity Domain and Projection Analysis ^[30]	√	x
Analysis of image chunked projection ^[31]	√	x
Local baseline detection and connectivity domain assignment ^[32]	√	x
Text line outline tracking ^[33]	√	x

Table 1 above makes clear that accurately detecting or situating identified characters is the fundamental task of character recognition in Tibetan historical literature, and that inaccurate character detection significantly affects recognition. Baseline estimate, connection domain analysis, and projection form the foundation of most text line segmentation techniques used in Tibetan historical sources. The segmentation of a single character can be done flawlessly by vertical projection as long as the text line is positioned accurately in the early stages. Zhao Dongcai, in conclusion, presented a BP network-based Tibetan single-character recognition technique for old Tibetan woodcuts [34]. Zach Rowinski developed the Namsel OCR system in 2011 [35]. It is capable of reading orthodox Tibetan books and contemporary printed Tibetan documents, but it is unable to recognize materials with adhesive characters, such as handwritten copies and historical Tibetan woodcuts. The work moved to UC Berkeley in 2013, when Rowinski started collaborating with other academics to further develop the system's capabilities. He also integrated Heyadati's early work on the hidden Markov model and actively collaborated with the Tibetan Buddhist Resource Center to produce a significant quantity of Tibetan electronic texts. Source [36] suggests a technique for recognizing Tibetan characters that does not require a lot of training samples and is based on component position information.

Based primarily on conventional techniques, character recognition research methods in Tibetan historical books published both domestically and internationally are based on the body of existing literature. The majority of the central algorithms in the aforementioned research are conventional pattern classification techniques, which typically require cutting text lines first, followed by characters, and then forwarding them to a single character recognition module for identification. Traditional machine learning techniques are used to investigate it, and it often focuses on topics like character segmentation, feature extraction, and classification algorithms. The early character segmentation benefits and

drawbacks have a direct impact on the recognition effect. As deep learning technology has advanced and processing capacity has increased, deep learning-based techniques have progressively gained traction and are now more effective and accurate at recognizing. Therefore, studies on the deep learning-based text recognition technique for Tibetan historical manuscripts have drawn attention. The next part will include specific research progress reports.

3.2 Deep learning-based text recognition of Tibetan historical documents

To start, there is a chance of mistake superposition and the entire procedure is laborious. Furthermore, certain empirical processes are necessary to determine the thresholds. Finally, the semantic association information of the text area context is completely ignored by the standard technique. As a result, an increasing number of academics are focusing on the deep learning-based approach to text recognition of Tibetan historical documents, with text detection and recognition being the primary responsibilities. Figure 1 illustrates the intricate and diverse structure and font shape of Tibetan historical documents, showcasing strong character adhesions both within and across lines within the text space. It is nearly hard to create a universal system with great generalization using traditional document analysis and recognition algorithms because they demand inspirational parameters. Figure 4 illustrates the precise distinction between deep learning techniques and conventional approaches.

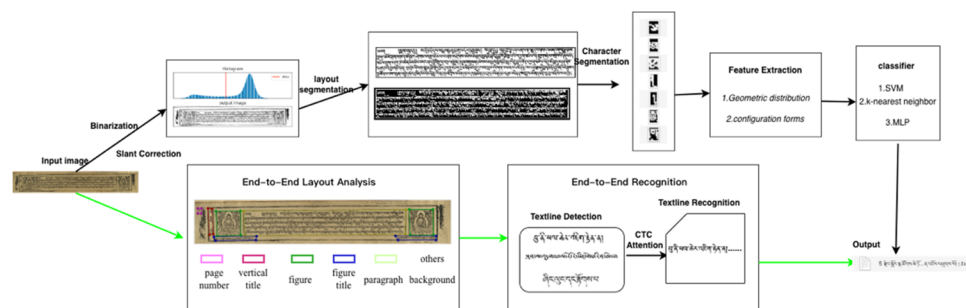


Figure 4. Comparison of the Two Methods' General Processes for Text Recognition of Tibetan historical document images.

(1) Tibetan historical documents image text detection: The method used to find text in an image is called text detection. Text recognition relied mostly on manual feature extraction prior to the widespread use of deep learning techniques. The field of text identification has benefited considerably from the advancement of deep learning, particularly in the area of object detection. Naturally, text detection differs from general object detection in certain ways. The trend of picture text line detection in historical Tibetan texts is reflected in the study that is now available. As an illustration, Wang Mengjin presented two deep learning-based picture text detection models for Tibetan historical texts in his master's thesis. These models were developed using the CNN+RNN and FCN+NMS frameworks, respectively, and showed good training effects on bespoke datasets [37]. A large-scale text line detection dataset of historical Tibetan writings was created in Ref. [38] to address the issue of the absence of publicly available datasets in this field. The dataset's efficacy was confirmed by tests using three scene text detection algorithms: PSENet, PixelLink, and DB. The unique construction technique has reference value. Ref. [39] proposed a text detection method based on a pixel-level semantic segmentation network architecture for Tibetan historical document images. Zhao Penghai utilized SOLOv2 to accomplish the layout analysis, the U-Net model for the picture binarization of historical Tibetan woodcuts, and Vision Transformer to finish the text line identification [40]. A CTPN-based methodology for identifying the longitudinal and transverse features of old Tibetan woodcuts was presented by proprietor Rinchen [41]. Although the training dataset for the system lacks images of historical Tibetan texts, experimental results indicate that the technique performs well [42].

(2) Tibetan historical document image text recognition: Character recognition is a system that uses a location-based text image and turns it into a computer text code. Character segmentation and recognition are involved in text recognition, which primarily refers to the identification of text lines. Its primary foundations include sequence learning machine learning theories and methods, as well as general pattern categorization or pattern recognition. The application situations, document picture style, and document attributes are intimately linked to the particular approach. Since 2019, In order to achieve text recognition of photographs of Tibetan historical texts, academics have gradually embraced deep learning techniques. Han Yuehui identified the photos of Tibetan history texts retrieved based on baseline estimation in the same year [43] using a CNN model. For Tibetan historical literature, Ref. [44] presented a CNN-based single-character recognition model that can identify 562 frequently used Tibetan characters. For historical Tibetan woodcuts, Hu Pengfei presented an end-to-end CRNN-based text line identification approach [45]. Renqing Dongzhu primarily created a text

line recognition model for Tibetan historical woodcuts with severe character adhesion between lines using residual networks and bidirectional long short-term memory recurrent neural networks. This model successfully addressed the recognition and correction problem of similar Tibetan characters during the post-processing stage[41]. In Ref.^[46], a method known as MbSPPVGG—a multi-branch Deep Convolutional Neural Network (CNN) and Spatial Pyramid Pool (SPP)—was presented for character identification of historical Chinese ethnic minority writings. This method primarily addresses the issue of multi-class character recognition with background noise and damaged images of history Chinese ethnic minority documents, including Tibetan. After that, several researchers combined the knowledge-driven model with the data-driven supervised learning method in an attempt to narrow down the search space in the Tibetan character identification stage. They then developed a convolutional structure called TwinNet for the recognition of comparable Tibetan characters^[47]. LeNet, VGGNet, ResNet, and Wide-ResNet models were employed by Ref.^[48] to classify historical Tibetan characters. It was discovered that the Wide-ResNet model, with an accuracy of almost 94%, had the best classification effect on Tibetan single-character images. Ref.^[49] examined the detection of multi-text lines in unsegmented Tibetan document pictures using the ResNet+CTC framework. A full-page technique for the accumulation of errors in end-to-end text recognition was proposed by Yang Xiaolong et al.^[50].

3.3 Related datasets

As of right now, the only dataset available in this subject is the NMU-THDTL Tibetan history book dataset(<https://pan.baidu.com/s/1sgScgacEcJzHDDXZHjWktQ>), which was released in Literature^[22] and is appropriate for the purpose of segmenting Tibetan historical literature. As a result, some academics have put up several ideas to address the dearth of pertinent datasets. For instance, Zhou Fengming created a sample database of 5,000 sets of ancient Tibetan woodcuts in Ujin script and created a character collecting system that gathers 7,240 Tibetan characters (Basic Set, Extended Set A, and Extended Set B) on the PC platform^[51]. Renqing Dongzhu et al. created a training dataset for character recognition of woodcuts in old Tibetan books using an image synthesis technique, and they discovered through experimentation that the dataset was useful^[52]. Using photographs of authentic Tibetan historical texts, Ref.^[53] extracted Tibetan strings to create a dataset with 610 different types of Tibetan characters.

3.4 Application of image text recognition in Tibetan historical documents

The development of artificial intelligence, machine learning, and other related technologies has led to the creation of several Tibetan OCR software, such as TH-OCR, which was developed in collaboration with Tsinghua University and Northwest University for Nationalities; UTibet-OCR-2008, which was created by the Engineering Research Center for Tibetan Information Technology of the Ministry of Education of Tibet University (downloadable packages particular to this software may be found in reference [41]); Tesseract (<https://groups.google.com/d/msg/tesseract-ocr/ONkAD2kuxUQ/EQsepM67D94J>), and Abbyy OCR. The software has deficiencies in its capacity to identify Tibetan characters that have been converted into Sanskrit.

The aforementioned Tibetan OCR software is intended for the modern printed version of Tibetan documents. This year, the Tibetan University's Ministry of Education, Engineering Research Center, produced two types of Tibetan historical book text recognition systems: the woodcut book layout analysis and text recognition system available at (<http://bmf.utibet.edu.cn/socr.fds>) and the handwritten book layout analysis and text recognition system available at (<http://bmf.utibet.edu.cn:9000/socr.fds>). They require more improvement in identification accuracy, and their anti-interference capabilities is weak. Additionally, it's said that Google has integrated OCR software for Tibetan historical literature into its Google Books and Google Drive products. This work is still in progress, but initial accuracy has approached or beyond 90%.

In conclusion, there is a need for immediate improvement in the difficult research field of intelligent character identification of Tibetan historical writings.

4. CONCLUSION AND OUTLOOK

A growing number of academics are interested in character identification in Tibetan historical books as a result of digitalization of historical books, national strategic needs for safeguarding historical cultural resources, and advances in artificial intelligence technology. in order to provide interested investigators with a specific reference.

The image text recognition of Tibetan historical documents still confronts numerous obstacles because of the various layouts of different editions of Tibetan history books and the unequal picture quality of Tibetan historical documents.(1) Lack of data: using deep learning algorithms is greatly hampered by the lack of public datasets that include the whole

layouts of Tibetan history books, uniform distributions of Tibetan characters, Sanskrit transcriptions of Tibetan characters, and large amounts of data that are publicly available. (2) Difficulty in preprocessing pictures of Tibetan historical documents: most Tibetan historical books are smeared, damaged, and deformed, and have a range of engraving or writing techniques that cause character adhesion and make it impossible to define Tibetan syllabic dividers. As a result, these properties require that the image enhancement and binarization methods be optimized; (3) Character adhesion makes it harder to distinguish text rows and columns because there are many adhesion and overlapping characters, particularly syllabic separators and single vertical characters that overlap with neighboring characters, and there are little spaces between lines of characters in Tibetan historical writings. (4) There are several kinds of Tibetan characters at the D level. There are up to 1,729 different types of Tibetan character encoding standards, referred to as basic sets and extended sets A, according to the national standard GB29274-2012, which was released in 2012. The Tibetan character set, known as Extended Set B, was significantly increased in the same year that the national standard GB29278-2012 was established; this standard also specified the 5701 Tibetan encoding standard, which is mostly made up of Tibetan characters and Sanskrit transliterated Tibetan characters; (5) There is still a lack of sufficient precision and reliability in Tibetan recognition: the primary causes are the numerous comparable yet confusing characters in Tibetan, the inadequate diversity and variety of training data, and the poor early character detection performance; (6) Inadequate capacity for model generalization: Character recognition algorithms for Tibetan historical books now in use are typically trained using samples from a predefined set of categories; as a result, untrained categories cannot be identified in real-world scenarios, and there is a severe forgetting issue for categories with few data. (7) An excessive reliance on big data: The current research on character identification of Tibetan history books is reliant on a huge amount of manually annotated data; (8) The application of mainstream deep learning algorithms is inadequate; from small sample learning, semi-supervised learning, weakly supervised learning, self-supervised learning, open set recognition, and incremental learning to increase the learning ability of Tibetan historical book character recognition models. In spite of these efforts, there is still a shortage of data synthesis techniques for high-fidelity Tibetan historical book samples. To find fresh research problems and answers, keep a watch on the most recent research trends in real time.

Generally speaking, research on other historical book versions has not yet been included in the current stage of Tibetan historical book character identification, which is primarily focused on Tibetan historical book woodcuts. The performance evaluation of various recognition models is hampered by the absence of accessible datasets, which also restricts research and development in this area. Furthermore, one of the key research challenges in the future will be how to better utilize unlabeled actual data and align the model with real-world circumstances.

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REFERENCES

- [1] Weilan Wang. Modern Tibetan Language Recognition [C]// Chinese Information Society. 1998 Proceedings of the International Conference on Chinese Information Processing. 1998: 388-391.
- [2] Binmakhashen G M and Mahmoud S A. 2019. Document layout analysis: A comprehensive survey. *ACM Computing Surveys*, 52(6): Article No. 109.
- [3] Chen X, Jin L, Zhu Y, Luo C and Wang T. 2021. Text recognition in the wild: A survey. *ACM Computing Surveys*, 54(2): Article No.42.
- [4] Chenglin Liu, Lianwen Jin, Xiang Bai, Xiaohui Li, Fei Yin. 2023. Frontiers of Document Intelligent Analysis and Recognition: Review and Outlook. *Chinese Journal of Image and Graphics*, April 7, 2023.
- [5] Yong Cuo, Nyima Tashi, Zhengzhen Liu, Qiuhua Wei, Luosang Gadeng, Gama Trashi. Layout analysis of Tibetan historical documents based on deep learning[J]. *ACM International Conference Proceeding Series*, 2019. 96 101.
- [6] Mengjin Wang, Yongcuo, Li Shanchen. Current status of research on text detection in Tibetan historical books [J]. *Computer Knowledge and Technology: Academic Edition*, 2020, 16 (10): 4.
- [7] Xianba. Research on Tibetan Historical Book Versions - Focusing on Tibetan Historical Book Prints [J]. *Xizang Research* 2016 (3): 99-110+2+121.

- [8] Chongxian Zhu. Research on the Compilation of Chinese Ethnic Minority Historical Books and Documents [M]. Commercial Press, 2017.
- [9] Kojima M, Kawazoe Y, Kimura M. A study of character recognition for Wooden Blocked Tibetan manuscript[J]. 2016.
- [10] Hedayati F, Chong J, Keutzer K. Recognition of Tibetan wood block prints with generalized hidden Markov and kernelized modified quadratic distance function[C]//Proceedings of the 2011 Joint Workshop on Multilingual OCR and Analytics for Noisy Unstructured Text Data. 2011: 1-14.
- [11] Han Y, Wang W, Wang Y. Research on automatic block binarization method of stained Tibetan historical document image based on Lab color space[C]//2nd International Forum on Management, Education and Information Technology Application (IFMEITA 2017). Atlantis Press, 2018: 327-338.
- [12] Han Y, Wang W, Liu H, et al. A combined approach for the binarization of historical Tibetan document images[J]. International Journal of Pattern Recognition and Artificial Intelligence, 2019, 33(14): 1954038.
- [13] Hu J, Xiaowu L, Qiao S, et al. Geometric correction method for Tibetan woodcut document images[J]. Multimedia Tools and Applications, 2022, 81(11): 15609-15632.
- [14] Zhang X, Duan L, Ma L, et al. Text Extraction for Historical Tibetan Document Images Based on Connected Component Analysis and Corner Point Detection[C]//CCF Chinese Conference on Computer Vision. Springer, Singapore, 2017.
- [15] Xiuyou Wang, Zhengzheng Ren, Huaming Liu, et al. Segmentation of Tibetan Historical Books Layout Based on Segmented Projection Method [J]. Journal of Chifeng University: Natural Science Edition, 2021, 37 (8): 5
- [16] Fangzhen Ren, A Study on the Page Segmentation Method of Tibetan Historical Books [D]. Fuyang Normal University, 2021.
- [17] Yanxing Li, Research on Text Line Segmentation Method for Tibetan Historical Documents Based on Rules and Learning [D]. Beijing University of Technology, 2018.
- [18] Li Y, Ma L, Duan L, et al. A text-line segmentation method for Tibetan historical documents based on baseline detection[C]//Computer Vision: Second CCF Chinese Conference, CCCV 2017, Tianjin, China, October 11–14, 2017, Proceedings, Part I. Springer Singapore, 2017: 356-367.
- [19] Wang Y, Wang W, Li Z, et al. Research on text line segmentation of Tibetan historical documents based on the connected component analysis[C]//Pattern Recognition and Computer Vision: First Chinese Conference, PRCV 2018, Guangzhou, China, November 23-26, 2018, Proceedings, Part III 1. Springer International Publishing, 2018: 74-87.
- [20] Hu P, Chen Y, Hao Y, et al. Text line segmentation based on local baselines and connected component centroids for Tibetan historical documents[C]//Journal of Physics: Conference Series. IOP Publishing, 2020, 1656(1): 012034.
- [21] Li Z, Wang W. Tibetan historical document recognition of uchen script using baseline information[C]//Tenth International Conference on Graphics and Image Processing (ICGIP 2018). SPIE, 2019, 11069: 997-1006.
- [22] Li Z, Wang W, Chen Y, et al. A novel method of text line segmentation for historical document image of the uchen Tibetan[J]. Journal of Visual Communication and Image Representation, 2019, 61: 23-32.
- [23] Wang Y, Wang W, Cai Z. Baseline detection based on local connection and evaluation for Tibetan historical document text line[C]//Third International Conference on Computer Science and Communication Technology (ICCSCT 2022). SPIE, 2022, 12506: 1597-1603.
- [24] Wang Y, Wang W, Li Z, et al. Research on text line segmentation of Tibetan historical documents based on the connected component analysis[C]//Pattern Recognition and Computer Vision: First Chinese Conference, PRCV 2018, Guangzhou, China, November 23-26, 2018, Proceedings, Part III 1. Springer International Publishing, 2018: 74-87.
- [25] Ce Zhang, Weilan Wang. Character Segmentation of Wujin Tibetan Historical Books Based on Structural Properties [J]. Progress in Laser and Optoelectronics, 2021, 58 (20): 260-275
- [26] Li Z, Wang W. Tibetan historical document recognition of uchen script using baseline information[C]//Tenth International Conference on Graphics and Image Processing (ICGIP 2018). SPIE, 2019, 11069: 997-1006.
- [27] Zhang C, Wang W, Liu H, et al. Character Detection and Segmentation of Historical Uchen Tibetan Documents in Complex Situations[J]. IEEE Access, 2022, 10: 25376-25391.
- [28] Zhao Q, Ma L, Duan L. A touching character database from Tibetan historical documents to evaluate the segmentation algorithm[C]//Pattern Recognition and Computer Vision: First Chinese Conference, PRCV 2018, Guangzhou, China, November 23-26, 2018, Proceedings, Part IV 1. Springer International Publishing, 2018: 309-321.

- [29]Rumen Shi,Line segmentation of Tibetan historical texts based on A * and connected regions [D]. Fuyang Normal University, 2022.
- [30]Jincheng Li, Xiaojuan Wang, Weilan Wang, et al. Line segmentation of Tibetan historical texts based on text core regions and expanded growth [J]. Progress in Laser and Optoelectronics, 2021,58 (02): 113-123.
- [31]Ma L, Long C, Duan L, et al. Segmentation and recognition for historical Tibetan document images[J]. IEEE Access, 2020, 8: 52641-52651.
- [32]Hu P, Wang W, Li Q, et al. Touching text line segmentation combined local baseline and connected component for Uchen Tibetan historical documents[J]. Information Processing & Management, 2021, 58(6): 102689.
- [33]Zhou, Fengming and Wang, Weilan and Lin, Qiang. A Novel Text Line Segmentation Method Based on Contour Curve Tracking for Tibetan Historical Documents[C]// International Journal of Pattern Recognition and Artificial Intelligence,10(32):18540253.
- [34]Dongcai Zhao. Research on woodcut Tibetan scripture text recognition based on BP network [J]. Microprocessor, 2012,33 (05): 35-38+43.
- [35]Rowinski Z, Keutzer K. Namsel: An optical character recognition system for Tibetan text[J]. Himalayan Linguistics, 2016, 15(1).
- [36]Han Y, Wang W, Wang Y, et al. Research on the method of Tibetan recognition based on component location information[C]//Pattern Recognition and Computer Vision: First Chinese Conference, PRCV 2018, Guangzhou, China, November 23-26, 2018, Proceedings, Part III 1. Springer International Publishing, 2018: 63-73.
- [37]Mengjin Wang,Research on Text Detection of Tibetan Historical Books Based on Deep Learning [D]. Xizang University, 2020.
- [38]Zhi X, Gao D, Zhao Q, et al. Text Detection in Tibetan Historical Books: A Benchmark[C]//2021 IEEE 2nd International Conference on Pattern Recognition and Machine Learning (PRML). IEEE, 2021: 254-259.452-456.
- [39]Gong Quzhuome, Cai Rang Jia, San Zhi Jia. Text Region Detection of Tibetan Historical Books Based on Semantic Segmentation [J]. Computer Simulation, 2022 (005): 039.
- [40]Penghai Zhao, Layout Analysis and Recognition System for Wujin Tibetan Documents [D]. Northwest University for Nationalities, 2022.
- [41]Renqing Dongzhu Research on character recognition of historical Tibetan woodcut books based on deep learning [D]. Xizang University, 2021.
- [42]Xu Z, Zhu J, Liu Y, et al. Research on Arbitrary Shape Tibetan Text Detection with Graph Network[C]//2022 International Conference on Image Processing, Computer Vision and Machine Learning (ICICML). IEEE, 2022.
- [43]Yuehui Han, Design and Implementation of a Tibetan Historical Book Recognition System [D]. Northwest University for Nationalities, 2019.
- [44]Han Y, Hao Y, Wang W. Research on the character recognition of Tibetan historical document based on CNN[C]//Journal of Physics: Conference Series. IOP Publishing, 2019, 1237(3): 032052.
- [45]Pengfei Hu, A deep learning based recognition system for Wujinti Tibetan historical books [D]. Northwest University for Nationalities, 2021.
- [46]Guo H, Yang D, Liu Y, et al. Script identification of ancient books by Chinese ethnic minorities using multi-branch DCNN and SPP[J]. Pattern Analysis and Applications, 2023, 26(2): 809-821.
- [47]Hao Y, Wang W, Lin Q, et al. TwinNet: A Symmetrically Structured Convolutional Architecture for Identifying Tibetan Characters[J]. 2022.
- [48]Zhao Z. Research on single-character image classification of Tibetan historical books based on deep learning[C]//2022 3rd International Conference on Computer Vision, Image and Deep Learning & International Conference on Computer Engineering and Applications (CVIDL & ICCEA). IEEE, 2022: 1-5.
- [49]Liu Y, Zhu J, Xu Z, et al. Research on Multi-line Recognition Algorithm for Tibetan Document[C]//2022 3rd International Conference on Pattern Recognition and Machine Learning (PRML). IEEE, 2022: 72-76.
- [50]Xiaolong Yang, Hongmei Gao, Dingguo Gao. Research on Full Page Text Recognition of Tibetan Dunhuang Fonts [J]. Electronic Technology and Software Engineering, 2022 (16): 221-227.
- [51]Fengming Zhou, Design and Construction of a Sample Library of Wujin Sanskrit Tibetan Historical Books Character Ding [D]. Northwest University for Nationalities, 2018.
- [52]Ren Qing Dong Zhu, Tou Dan Cai Rang, Nima Zhaxi. Research on the synthesis method of Tibetan historical book character recognition training data [J]. Plateau Science Research, 2021,5 (03): 84-91.
- [53]Zhang C, Wang W, Zhang G. Construction of a Character Dataset for Historical Uchen Tibetan Documents under Low-Resource Conditions[J]. Electronics, 2022, 11(23): 3919.